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Andrew Abbott ^a

^a Department of Sociology, Rutgers University

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Conceptions of Time and Events in Social Science Methods

Causal and Narrative Approaches

Andrew Abbott
Department of Sociology
Rutgers University

There are two sides to the question of why people have certain kinds of careers. We can focus on *why* people have certain kinds of careers or we can focus on why people have *certain kinds of careers*. That is, we can worry about causality on the one hand or narrative typicality on the other.

The difference is paradigmatic. Those worried about causality see no point in studies that don't discover causes. Those worried about typicality see causes as so much reification. The different views entail assumptions that make them mutually invisible. They are like two different paths through the same park; one sees the same things from each, but those things look very different.¹

The contrast of causal and narrative analysis of course goes beyond the issue of careers. However, careers provide an empirical referent for an otherwise abstract discussion. Careers are a particularly good example because they mix chance and determinism. Stage processes, although widely theorized in social science, are usually conceived to be more regular than careers; the typical pattern is taken for granted, and we seek the causes that propel it. By contrast, interactional processes, although equally well known, are conceived to be looser than careers. There the focus is more on contingent developments, and typicality is presumed quite unlikely. In careers, we expect both a fair amount of pattern and a fair amount of fluctuation. Thus, they offer a particularly good example of the tradeoffs between thinking about historical processes in terms of causality and thinking about historical processes in terms of narrative pattern.

I will begin by drawing a general portrait of the two different ways of imagining careers and then consider the assumptions of these two views, focusing on major differences. These differences concern (1) the nature of the social process, (2) the working of causes, (3) the character and "orderability" of events, and (4) the nature of time. In conclusion, I will consider what these differences mean for methodological development.

TWO CONCEPTIONS OF CAREERS

There are two formal ways to conceptualize careers. On the one hand, one can treat them as realizations of stochastic processes.² On this argument, there exists some sort of underlying process with certain kinds of parameters. These parameters may be determining causes, such as an individual's race or education, or they may be preferences that dictate an individual's choices.³ This stochastic view sees the career as a realization: a career is simply the list of results that the underlying process throws up over succeeding time periods. It is, in that sense, an accident rather than a pattern, an appearance rather than a reality. Reality lies at the level of the underlying process of causes or choices.

As one might expect, career-analysis methods that derive from this view focus on outcomes at particular points. For example, the Wisconsin status attainment model assumes that a set of variables predicts occupational outcome at a point, the time point of the dependent observation with which the path model ends. The much more sophisticated durational models based on event-history data adopt the same idea but predict outcome at a succession of points. Again, there is no conception of the career as a whole. Some authors distinguish classes of cases that follow different sequences of outcomes and then seek differences in causal parameters between those classes. Such models remain within the stochastic framework. Sequences of outcomes—for example, event "careers" of people entering the labor force—are used merely to separate subgroups; there is no idea that the causes themselves might be subject to differing orders in differing cases within the groups.⁴

Alternatively, one might conceptualize careers as wholes. That is, one could view a career as a single unit at its completion. The varying impacts of opportunities, constraints, causes, and choices are all then taken to merge their effects indistinguishably in a single thing, the

career line. In this view, the career is a reality, a whole, not simply the list of successive realizations of an underlying stochastic process. It is important to recall that this is our commonsense construction of the history of persons and of many supraindividual actors. Thus, we consider academic careers to take one or another form. We make psychological predictions on the basis of particular life-event sequences. We talk about patterns in riots, revolutions, and other forms of collective behavior. Each of these discussions assumes that past careers—of individuals, social movements, organizations—can be treated as wholes, as units. Beginning, middle, and end are all of one piece, the one leading ineluctably to the next.⁵

However, methods for analyzing careers as whole units are rare. Peter Abell has argued for a homomorphic analysis that reduces such narratives to a number of common shapes. It is not yet clear how Abell's methods would be applied empirically. Optimal matching techniques seem to be the only empirically practicable techniques available. They measure the resemblance of career sequences by counting the numbers of changes required to turn one sequence into another.⁶

There are thus two ways of seeing careers, indeed two ways of seeing historical processes more generally. One focuses on stochastic realizations and aims to find causes; the other focuses on narratives and aims to find typical patterns. This dichotomy holds as clearly in other fields of research as it does in the study of careers. Thus, one can imagine revolutions as the realizations of stochastic processes, in which case the history of a given revolution is actually just the listing of successive outcomes of some underlying causal processes. On the other hand, one can see revolution as having a complete implicit logic running from start to finish, in which case the history of a given revolution is a logical narrative with an inherent telos. Similar arguments apply to life courses, organizational histories, occupational development, and so on.

MAJOR DIFFERENCES IN ASSUMPTIONS ABOUT TIME AND HISTORICAL PROCESSES

These two views of careers make a variety of assumptions about time and the embedding of social life in time. Of course, any particular author makes a diverse choice of these assumptions, but reflecting about two polarized versions of the assumptions makes one more aware of the choices that are made. If the following argument seems overly schematic, then, one reason lies in its desire to excavate our views of the social process. To a certain extent, however, I have made a polemical point as well. The stochastic view is overwhelmingly dominant in empirical social science. I wish to argue that, despite its real attractions, that view is not the only reasonable way to formalize social processes.⁷

Constitutive Assumptions

Assumptions about the social process begin with constitutive assumptions about the nature of social actors and their world. That is, the first assumptions of the two views are less about time passing per se than they are about the social beings involved in that time.

In the stochastic or causal view of careers, as in the broader "general linear reality" that I have analyzed elsewhere, the social world is made up of fixed, given entities with variable properties. We usually call these entities "cases" and their properties "variables."⁸ A career consists of the succession of the values of a dependent property or properties over time. The stochastic view aims to find the minimum number of properties necessary to generate these observed "careers," which are in fact most often considered one time point at a time, as in status-attainment models.

Such a procedure builds up career from a minimum of mere existence. Its paradigms are the literatures of industrial reliability and epidemiology. In these fields, existence itself is the dependent, and sometimes the only, property of interest; waiting time till death may, for example, be a function only of prior lifetime.⁹ Of course, the stochastic view of careers can envision considerably more complex matters, but it does so by adding more properties—more independent and dependent variables—to this foundation of mere being. It is striking that initiative and action in such "career" models belong to the variables, not to the cases. Variables do things; the cases themselves merely endure.¹⁰

By contrast, in the narrative or typical-pattern view of careers, the social world is made up of subjects who participate in events. The very language used to describe the position is radically different; "event" here means something quite different from "realization of a Bernoulli process." The subjects and events of the "narrativist" are inherently complex. Analysis occurs by directly simplifying them.

Rephrasing the narrative position in the terminology of the stochastic view helps to make the two views commensurable. In that terminology, what the narrativists call "an event" may be defined as a combination of particular values of many variables. One moves from the stochastic state-space to the narrative list of "events" simply by creating a list of neighborhoods (for a continuous state-space) or combinations of properties (for a discrete one) that exhaustively cover (in the continuous case) or classify (in the discrete one) the state-space. This is the list of possible events.

To a narrativist, the justification for thinking directly in terms of these events, rather than in terms of the dimensions or the uncrossed categories themselves, is the surprising empirical fact that once we consider a state-space of any real complexity, most events are null. That is, most combinations of particular values of discrete var-

ables are never observed, and most neighborhoods in a continuous variable space are empty. If it is the case that the state-space is mostly empty—if most possible events don't happen—why should we design models covering all possible events? Such models (general linear models, for example) try to explain not only what did occur, but also lots of things that didn't or perhaps can't (because of constraints) occur. It is more parsimonious to define the actually observed locations as "events" and to investigate their narrative structure directly, without recourse to reified "causes." We usually justify the use of "variables" on the ground that reality is too complex to do otherwise. Maybe that isn't true.

An empirical example is useful to illustrate the assertion that most events don't occur. I have randomly taken four variables on U.S. states from the *Statistical Abstract*: current expenditures per school pupil, crime rate per 100,000 population, percent of eligibles voting for U.S. representatives in the last election, and energy consumption (BTUs) per capita. To provide a strong illustration of "emptiness," I have chosen variables that were not correlated, for, of course, correlation would empty large areas of the space. (The highest correlation among these four variables, in absolute value, is .268.) To focus only on the relevant portion of the state-space, I have located each state (fifty plus the District of Columbia) in a normalized four-dimensional rectangle by rescaling the variables. I have subtracted the respective minimum from each variable and divided the result by the variable's range. This yields a cloud of 51 points in a four cube with unit edges. We can divide each dimension of this cube into two or more parts and then consider how many of the resultant "grid cubes" have a case or cases in them.¹¹

Because there are four variables, there are 16 grid cubes (or cells) if we divide each dimension in two and in general n to the fourth power cubes for any " n -cut" of each di-

mension. The data in table 1 show how many of the cells have specified numbers of occupants in this data. The number of empty cells is surprising. If we cut each dimension in two halves (a "two-cut") 7 of 16 cells are unoccupied. Overall, of course, the number of empty cells reflects the disparity between the 51 cases and the hundreds or thousands of grid cells. But note the concentration in the cells that *are* occupied. In a two-cut, 45 of the 51 cases are in 4 of 16 cells. In a three-cut, 28 of the cases are in 5 of 81 cells. In a four-cut, 23 of the cases are in 6 of 256 cells. Note, too, that one must divide this space into 10,000 cells to get each of the 51 cases in its own cell. Some of this clumping is random. But Poisson fits to this data show distinct evidence of further clumping, of real order. In a three-cut, for example, a Poisson model predicts that about half the cells should be empty; in fact, three quarters are empty—distinct evidence of clumping in other cells.¹²

The concentration shown in all these measures reflects local order unmeasured by global measures like the correlation coefficient. Even when these global measures show minimal relations between "variables," most of the state-space is empty because there are, in fact, many local resemblances between cases. The three-cut—its 81 cells being of the same order of magnitude as the 51 cases—provides the fairest evidence. Only 19 of these cells are occupied, and over half the cases are in 5 cells. A purely typological approach to this data could cover much of the data with five simple descriptions.¹³

The narrative analyst therefore views events as the natural way to simplify the social process. Rather than disassembling complex particulars into combinations of supposedly independent variable properties (i.e., rather than locating the cases in a multidimensional or crossed-category state-space at all), such an analyst views direct conceptualizing of the observed events as the best way to sim-

TABLE 1
Number of Cells with n Occupants

n	Subdivide each dimension by								
	2	3	4	5	6	7	8	9	10
0	7	62	225	587	1,253	2,355	4,047	6,511	9,949
1	4	10	22	30	36	42	47	49	51
2	1	4	3	4	6	3	2	1	
3			3	3	1	1			
4			1	1					
5		2	2						
6		1							
7	1	1							
9	1								
10		1							
11	1								
18	1								

plify the complex flow of occurrences. Rather than building up its cases by assigning them minimal properties beyond a foundation of existence, the narrative view assumes a complexity from which it must “build down,” or simplify. Existence, of course, still determines the ultimate beginnings and endings of careers, but it is inconceivable as a separate matter for inquiry.

A substantive example underlines the difference. Suppose we are considering the passage of workmen’s compensation laws in American states. To treat this problem from the stochastic view, one observes forty-eight states (the fixed entities) over some defined time period in which those laws are passed (for example, 1909–1929) and notes when they do and do not have compensation laws. The goal of the event-history approach is to find the minimal set of state properties (types of economy, productivity of manufacturing, etc.) that will predict when the laws will appear and when they will not. In the whole-career model, one establishes the basic events leading to workmen’s compensation laws in the various states and tries to figure out the typical sequence of events by which compensation laws come about. These events undoubtedly involve many of the “variables” used in the stochastic approach, but rather than treating them independently, the narrator will view particular combinations of particular values of those variables as events in one or more patterns for the compensation story. The narrative process is justified because with, say, fourteen variables, the vast majority of possible events must of necessity not be observed if we take the stochastic approach.¹⁴

Assumptions about Causality

These fundamental differences about how to simplify the complexities of social life are complemented by differences over how to understand causality. Ultimately, the two views accept the same ideas about explanation; both regard narration as the final form of social explanation. That the whole-career view does so is obvious, but stochastic writers usually appear to think that causality resides in variables. Yet careful reading shows that the language of “variables causing things” is merely a shorthand; stochastic writers fall back on stories or “plausible mechanisms,” when they must defend or support particular assertions about the variables. Narrative is the fundamental recourse.¹⁵

This ultimate common reliance on narrative is entangled with other causality assumptions that are quite different. A first assumption concerns the pacing of causes. Although not required to, stochastic models generally assume that causes work “at equal speed” across all cases. Whole-career models, by contrast, imagine causal pacing as potentially varying. Consider two careers in which the same sequence of jobs is observed, but in one of which the duration of each job is exactly twice that in the other. Any model examining transitions or probabilities

of motion will treat these durational differences as generating variance in estimates of the effect of the causal predictors on the careers. By contrast, if we consider the careers as a whole, we could propose that the causes are in each case the same, but that they are “working at different rates.” Within the stochastic view, this is an assertion that some unmeasured variable(s) interacts with the observed variables, since time itself presumably causes nothing. A whole-career approach allows us to see that interaction effect—and hence raise the issue of unmeasured variables—relatively easily. Moreover, we might wish to consider the actual idea that causes can work at different rates. This is a standard assumption of working historians, and, as we shall see below, assuming uniformity of pacing involves problematic assumptions about the continuity of causal effects. It is for this reason that differences in assumptions about causal pacing matter.

A second difference about causality concerns relevance. As I have argued elsewhere, the stochastic view must make the assumption that causes are always relevant; once in the model, they must remain. In formal terms, this means that any method employing linear transformations for modeling (whether directly or as an exponential term, as in partial likelihood methods) has to assume that the dimensionality of the transform involved doesn’t change. Because the whole-careers view looks at careers in terms of events, it can regard the causal factor of being black or female as affecting some of those career events but not others. There is no need to presume a perpetual importance. Of course, in the stochastic view this problem could be formally handled by allowing the coefficients of the underlying transformation to vary continuously and by assuming that most of them are zero most of the time. But that procedure throws away the benefits of assuming the transformation model in the first place and precludes estimation in most situations. It would be a very cumbersome means of handling a problem confronted directly by the whole-careers view.¹⁶

This difference is closely related to a more general difference between historiographical procedures and social science ones. Historians write their narratives to follow the causal action. If that action now seems to involve one or two principal causes and not other causes central just one or two events ago, then the historian simply ignores the present of those prior causes. They are assumed to be bubbling along in the background and not really affecting much. Such a view directly contravenes the views of most social scientists. The physical science model on which social scientists try to operate makes no allowances for causes that appear and disappear. One can hardly imagine a sociologist saying that “race simply didn’t matter much except at a couple of crucial conjunctures.” Yet this kind of argument is clearly implicit in standard narrative procedures in history.¹⁷

A third basic difference concerns the meaning of causes. The stochastic approach assumes, broadly speak-

ing, that the meaning of a given variable doesn't vary with historical time and with the context of other variables. Interaction (the formal term for the latter context) is basically assumed to be secondary; main effects are primary. By contrast, the whole-careers approach, with its focus on events, assumes that interaction comes first. For as I argued earlier, the whole-careers approach sees as events what the stochastic view would call constellations of particular values of variables, particular values of interactions. Main effects are a fiction with which the narrativist dispenses, because social life doesn't actually occur in main effects, but rather in events/interactions. Gender, the disembodied characteristic, in fact causes nothing to happen. Men and women cause things to happen. And men and women never lack other characteristics than their gender. Life never occurs in main effects.

This difference reflects the constitutive assumptions discussed earlier. The stochastic approach starts with cases whose only property is existence and adds to existence as few other properties (mostly main effects, but sometimes interactions) as can explain the course of the careers. If main effects suffice, they suffice. By contrast, the whole-careers view sees cases as inherently complex and uses typologies of events, cases, and narratives to simplify that complexity. Such typologies take complexity as primary, rather than deriving it as the product of crossed categorizations or dimensions, as would a "main-effects" approach.

This focus on interaction means rethinking how we assign causal meaning to "variables." In the narrative view, the meaning of a particular value of a "variable" is not fixed by its relation to other values of *this* variable among the other cases, but rather fluctuates with its status as one among the several values of different variables that make up various particular events. To take my earlier example of workmen's compensation laws, the impact of a given level of productivity in Massachusetts is not defined with relation to levels of productivity in other states but with relation to the conjuncture of productivity, industrialization, and other variables in Massachusetts at a particular time. Main effects as such don't exist.¹⁸

This same argument applies to temporal context. A particular value of productivity may have no absolute meaning independent of time. Of course, the stochastic view can approach this problem via change scores. But the temporal contexts of variables may be more complex than change relations can capture. A given value may acquire significance because it is the first reversal of a long, steady fall or because it initiates a long, steady state. In either case, it is the general temporal context, not the immediate change, that matters. Only when this preceding temporal context has a standard linear impact on the present can the stochastic view really handle temporal context, as in ARIMA methods, and even there the analysis of multiple variables is a central problem.

Assumptions about Events and Their Orderability

Although the stochastic approach to careers doesn't really operate in terms of events, it does include distinct assumptions about the ordering of values of variables. To show how these assumptions work, I must introduce the concept of a variable's "time horizon," a time within which one can observe a meaningful fluctuation in that variable. The idea of "meaningful fluctuation" assumes that any variable is subject to measurement errors or "minor variation" (the two are probabilistically equivalent here) but that we know enough about those errors or variations to be certain, after some quantity of change in the variable, that an observed change is not an error or a random fluctuation but a substantive change. Thus, the time horizon is the period it takes to separate signal and noise or real and random change.

Time horizons fluctuate from variable to variable. In some cases, they reflect the relation between the variance of the error process and the determinate coefficients of (say) autoregression. If a large error variance is superimposed on a slow determinate trend, it will take us longer to see the trend than if the superimposed error variance is small. This happens because of the practical strategy we use to identify changes in variables. Because we wish to avoid mistaking random variation for true trend, we set our criterion for "real change" in terms of the error distribution.¹⁹ To some extent, then, varying time horizons arise in measurement error and other forms of superimposed random variation. In this case, they are a practical problem but not a theoretical one.

Often, however, time horizons differ in substantive, theoretical ways. For example, time-horizon fluctuations also arise in relations of inclusion. The most obvious, but by no means the only, such example is the time-horizon difference between a macrovariable and its microconstituents. Intuitively, we think of aggregate measures as lacking ontological status in themselves; they are simply sums of individual-level measures, in which case they would seem to have the same time horizon as the individual-level measures. But in fact, we are likely to notice a change in one individual's job satisfaction faster than we can observe a change in the job satisfaction of a group of individuals. This is true for both conceptual and probabilistic reasons.

Theoretically, time horizons differ because emergents such as group job satisfaction do have reality and can't be conceptually reduced to the sums of individual-level measures that we use to indicate them. The job satisfaction of a group is a cultural construct with real consequences. Although we can measure it only by aggregating individual-level measures, clearly those are just indicators of the fuzzy thing itself. The causal consequences flowing from recognition of a social fact like "workers are dissatisfied with their conditions of employment" are quite different from the causal consequences flowing from thou-

sands of individual statements that "Jones hates her job." In the one case, we may have revolution; in the other, we have burnout and other forms of individual response.

The mathematical argument that aggregate measures have longer time horizons requires no such emergentist assumptions and is therefore—to those who don't like such assumptions—all the more telling. In formal terms, time horizon is the waiting time until a result exceeding a certain limit is passed, that limit being defined as some percentile—say, the ninety-fifth—of the distribution of the underlying variable. (For simplicity, I dispense with the signal/noise model and assume an underlying normal distribution.) Assume that we "move the bell over" so that it is centered at the old ninety-fifth percentile, and that the shape of the distribution does not change.²⁰ How many samples will we take before we notice what has happened? Half of the potential samples in the new distribution still lie below our criterion (the ninety-fifth percentile of the old distribution and the fiftieth percentile of the new one) and thus don't tell us that change has taken place. The distribution of the waiting time until we get one above the mean is geometric; we have a (.5) chance of success in the first trial, a (.5*.5) chance in the second, a (.5*.5*.5) chance in the third, and so on. The mean of the geometric distribution—the expected waiting time to the first success (i.e., to what we have decided to call a measurable difference)—is q/p , where q is the probability of failure and p that of success. In the present case, that value is (.5/.5), or one trial. We are likely to have to wait one trial to get a value above the mean of the new distribution, the ninety-fifth percentile of the old one.

Now consider the situation when we add together n situations exactly like this, that is, when we add measures on n different individuals together.²¹ The distribution of the sum of n such geometric variables is an n -fold convolution of geometric distributions. (Intuitively, convolutions pick up the idea that when we add two distributions any value of the summed distribution can arise in many possible ways.) The expected value of the n -fold convolution of the geometric distribution with itself is simply n times the expected value of the original distribution, in this case n times 1. That is the expected waiting time until we will observe a change in the summed (or averaged) variable. This time is substantially longer than in the case of individuals. Our notion of time horizon has, then, a sound probabilistic basis.²²

The concept of time horizons allows us to see important differences between the two approaches considered here. Stochastic methods must assume what is technically called partial ordering of the variables' time horizons. Causality can obtain between variables of equal time horizons, or it may flow from variables of longer time horizons to those of shorter ones. But it cannot flow the other way, for one could then envision an observation interval such that the short-term variable would exhibit meaning-

ful variation but not the long-term one. In such a case, we could not believe that the short-term variable causes the long-term one, because we can measure only error in the latter.

Another assumption crucial to the stochastic approach is that, generally speaking, succeeding observations of all variables are what I shall call contingently independent. That is, we assume that if there is autoregression, it takes a given pattern that does not vary through time periods. Values of a variable in succeeding time frames cannot be linked in arbitrary ways. Suppose we are estimating some dependent aspect of a job career with multiple-independent-variable regression done in time-series format; successive years constitute the "cases," and some variable aspect of career—income, for example—is predicted from a vector of potentially lagged independent variables. If there are arbitrary patterns linking successive variables of an independent variable, such as temporary work assignments that last several observation periods and that reflect "events" in the economy lasting several observation periods, the varying autoregression in the independent variables does complicated and unspecifiable (because interactive) things to our parameter estimates. The problem is that certain succeeding values of various independent variables are linked in "events" that span several observation frames.²³

The time-horizon problem is handled differently in the whole-career view. Here all that matters is the list of events. They can be big or small in time horizon. In certain kinds of careers, for example, having a certain kind of patron at a particular stage determines the pattern of the entire rest of the career. This is an immediately determinable event, as opposed to other events of longer time horizon such as "mastering fundamental professional techniques" or "holding various minor jobs." The only narrative assumption that need be made is that an event can affect only events beginning after it in the career.²⁴ Because the narrative moves from event to event, there is no necessity even to assume a regularly spaced observation framework, such as observation every year. A small, highly specifiable event can be seen as affecting large, diffuse events. The only necessary assumptions involve orderability.

The stochastic and whole-career views also think about the sequence of events somewhat differently. In the stochastic approach, the standard assumption is that the fundamental order is an order of variables and that all cases obey this same order. In any multiple regression, for example, it is assumed that the pattern of causality is the same in every case. This is equivalent, given the relation of stories about "causal mechanisms" to path diagrams, to the assertion that every case follows the same narrative. One can, of course, use dummy variables, or simply split up one's cases, to avoid this assumption to some extent. But the

characteristic aim is to find a pattern of causes that holds across all cases rather than to find a variety of patterns. A related assumption maintains that no particular sequence of particular values of variables has any particular causes or consequences. (Technically, this is part of the “main-effects-first” assumption.) That is, we can’t say that whenever we see particular events a,b,c in that order we will always observe d. No statements are made about particular values of a,b,c; statements are made only about variables A,B,C, and D.

In contrast, in the whole-careers approach the order of particular events is the center of interest. Our first aim is to consider whether there is one or several characteristic sequences of events. Here the assumptions involve our ability to place a set of events in order. One version of the assumptions assumes strict order; there is no overlap between events. Another may permit some overlap but place restrictions on its amount. Another strategy uses the basic observation framework of the stochastic method but defines the observations in terms of events: the joint event of being of such and such an age and hold such and such a job, and so on.

Because order is central to the whole-career approach, we must be able to define order rigorously for that approach to work. There seem to be two basic strategies. One retains time-horizon flexibility and defines events conceptually. It considers a reality of observable “occurrences” and imagines how these might be colligated into conceptual “events.” Occurrences are things like taking courses or holding particular jobs; events are things like “getting an education” and “developing as a professional.” Constraints on the orderability of the events can then be seen as issues of measurement given temporal patterns of occurrences within events. Alternatively, we may define events combinatorially within a standard measurement framework, as just suggested. This produces orderability automatically but at the price of losing all information about the temporal “shape” of events—their duration and their intensity in terms of producing occurrences—in short, their time horizon. Thus, this last approach is a hybrid of a stochastic conceptualization and a whole-career one.²⁵

The Character of Time

Finally, the stochastic and whole-career approaches differ with regard to the character and depth of causal time. The stochastic view implies a belief that the social process is continuous and causally shallow, whereas the whole-career approach sees it as discontinuous and causally deep. These are not, of course, conscious assumptions; rather, they are implicit in the ways that the two approaches address their problems.

The characteristic stochastic study of careers takes discrete repeated observations on its variable (i.e., panel data). If we are considering the effect of education at

time 1 on income at time 2 five years later, our first idea would be to regard education as a step function, measured by degree (high school, B.A., M.S., Ph.D.), and to estimate its effect across cases in the normal way. But it would obviously be preferable to have annual measures of education, because education is not simply a matter of degrees but also of on-the-job training, and the latter might have its own distinct effect on income. If our measurement frame were annual, we could see the causal effect of degree in year 1 as falling on some intermediate variable—for example, general occupation in year 2—and see that general occupation as providing access to other forms of on-the-job education that would then affect detailed occupational position in year 3 and so on. That is, the original education-income effect over five years would be regarded as interpolating a whole series of lesser processes. With those processes in the model, it seems unnecessary and perhaps even illegitimate to draw a causal arrow from education in year 1 directly to income in year 5.

It is this interpolation that makes us prefer more panels to less in longitudinal work and that makes the event-history data framework, with its exact dates of events, so appealing. More data is better. Better dating is better.²⁶

But in the limit, we have here the continuity of calculus. There is no time interval so small that independent variables are not affecting dependent ones within it, for if we continue unmasking the interpolative character of coefficients, we end up at daily or hourly measurement. However at this level, we invariably think of reality in particular narratives. In those narratives, general variables like “my education” are simply resources in interactional facework; their consequences for one’s income arise through the privileges or disadvantages that occur in interaction. And these daily narratives of real interaction generally don’t involve most of the variables of long-run importance, or if they do, they do so in infinitesimal ways indistinguishable, in any practical sense, from error. At such a fine temporal level, reality is quite discontinuous and choppy. Major changes (e.g., in job status) are likely to be quite abrupt; the record that justifies a major promotion may take years to build, but the promotion itself is instantaneous. That is, more panels are better, but only up to a point. If we look too closely, the effects of global variables such as “my education” become very hard to specify meaningfully.²⁷

This *reductio ad interactionem* presents the stochastic view of careers with a fundamental dilemma. The stochastic view’s main assumption is that the career is simply a sequential list of the results of an underlying and real stochastic or choice process. A set of initial values faces the model and produces new values that in turn face the model and so on; the immediate past is perpetually producing the future, much of it by reproducing constancy, of course. For such a model, it is obviously

better to have more iterations than fewer, and yet after a certain number of temporal subdivisions, the assumption that every variable (or its constituents) is always in the model obviously breaks down. One requires a "microtemporal model" that addresses these local discontinuities but that can somehow produce the appearance of a more continuous reality to be interpolated at the macrotemporal level.²⁸

The whole-career view, like any narrative view, dispenses altogether with this insistence on an absolutely continuous historical time. Events have finite duration. They begin and end. There is less of a microtranslation problem because larger historical time has the same discontinuity as does microtime. Particular causes are not always present, but matter only when they are part of the complex particulars the narrativist calls events. It should be noted that although this view has all the weight of historiography behind it, it is quite shaky philosophically. Mead, Whitehead, and many others have argued that reality flows from deep past to immediate past to present and future and that the deep past has causal effects only through having determined the immediate past. The stochastic view founds itself on this quite reasonable assumption. One's doctoral location, for example, may affect initial university placement, but its effects on later events should work through its effect on that initial placement.²⁹

Yet there are a number of reasons for considering such historical "action at a distance." One of them is implicit in Mead's own theory of the past. The present is continuously reshaping the past in line with its concerns. In career terms, people constantly reinterpret their past job history in order to decide what to do now. Suppose a period of bad market conditions has sent people to jobs and organizations that they would otherwise have ignored, and this period is then followed by better market conditions. Some may then redefine their years on the periphery as useful preparation for current changes, whereas others may view them as mere waiting time, and still others may have lost sight of their peripheral or undesired qualities altogether, perhaps seeing this period as one of liberation from a career treadmill. These different analyses have varying implications for the individuals' future choices, even though all the people involved objectively may have had the experience of going to (what they at the time thought to be) the periphery. Because people are constantly assigning new historical efficacy to far past events by reinterpreting their careers in light of them, we must consider the far past to have some sort of causal efficacy.³⁰

We must also assume that there is historical action at a distance because, as I have shown, it is difficult to do otherwise. Because insistence on the continuity of historical processes proves impractical, we must make do with the alternative view that events have finite and varying durations and that they may overlap. A career

may be simultaneously shaped by ongoing events that began at very different times in the past: overall shifts in labor demand consequent on changes in the division of labor, recent vagaries of recruitment, changes in the viability of individual hiring organizations, and so on. There is no need to reiterate these issues for those who have lived through the academic market since the 1960s. Note, however, that this position involves us in the assessment of multiple overlapping events.

CONCLUSION

In this article, I have aimed to distinguish two fundamentally different views of historical processes. According to one view, historical processes are generated by stochastic processes. The social world consists of individuals with particular properties. Causes work at fixed rates in all cases; are generally, rather than occasionally, salient; and primarily work independently, although at times combining in interactive effects. Causality flows to some extent from context to detail but generally between variables of "equal" status. There are few or no patterns of causality reaching beyond a single iteration of the model, and the patterns that are observed affect all individuals equally: everyone has the same causal history. Social time is essentially continuous.

The other view sees historical processes as whole stories. The social world consists of complex subjects to whom complex things—events—happen. Causality flows differently in different cases, perhaps at different rates, certainly in different patterns. Most causes work in complex bunches, the events, rather than alone. There is no necessity that these events be of a certain consistent temporal size or length and no restriction on relations between events of differing sizes. Events can come in a variety of temporal orderings: strict sequence, overlap, simultaneity. All of this means that historical processes are fundamentally discontinuous.

My first purpose in distinguishing these views has been to make them visible. The second is to establish the latter of them as a possible foundation for methods addressed to the question, What are the characteristic patterns in social narratives? Our present approach to social processes following the first approach has been productive; I am certainly not denying that. But it makes many assumptions that restrict its vision.

The central restrictive assumption is that most events happen; hence, the variety of historical processes is overwhelmingly great. This assumption implies that the best models are global models aimed at global regularities, for it implies that these are the only feasible models. The idea that complexity requires global models in turn entails highly restrictive further assumptions: uniform causal pacing, relatively consistent causal effects, main effects dominating interaction, consistency of time horizons, and single causal order across all cases. Thus, if we stop

believing that most events happen, we can open new possibilities for formal analysis of historical processes.

These new analyses will not instantly produce results comparable to those we presently have. One can use optimal matching, for example, to examine the development of welfare states and develop measures of resemblance between these several histories. However, causal interpretation of those resemblances presents problems. For example, to find what overall causes determine the contrast between one group of welfare histories and another, one has to solve the problem of when to measure the causal variables: as of the beginning of the histories? as of a particular fixed date across all histories? as of the midpoints of the histories? The answers are not obvious. In the short run, a shift to narrative formalism means forgoing much of our customary interest in causes. Of course one can answer that most of our causal results, based as they are on the preceding strict assumptions, are little more than polite fictions. But still, narrative methods will provide no easy answers.

On the other hand, narrative positivism has some distinct and powerful advantages. It will facilitate direct communication between history and the social sciences, because it thinks about social reality the way historians traditionally have done. It will provide us with a method for directly addressing questions of typical sequence(s) that are central to a number of contemporary empirical literatures: life course, organizations, labor markets, and revolutions. It will uncover regularities in social processes that can then be subjected to causal analysis of a more traditional sort. Eventually, it may more fully analyze the complex patterns of history than the causal methods are able to at their best. In the meantime, I think serious reflection about basic temporal assumptions can help us all improve our work.³¹

NOTES

This article was originally presented at the Social Science History Association Annual Meeting held in Washington, D.C., on November 17, 1989. I would like to thank members of that audience for various useful comments. The article summarizes arguments I am making at greater length in a book manuscript currently in preparation.

1. I speak from personal experience, having submitted formalized narrative analyses to journals whose reviewers are trained to think causally. Nonetheless, the usual disclaimers apply to the following paper. I am overdrawing distinctions in order to underline important choices. Few analysts will take uniformly consistent positions on the assumptions I shall analyze. And good work can be done from either perspective.
2. Hereafter, I will use the phrase "stochastic view" as a shorthand for this view. I do not mean to imply thereby that the alternative "whole-career" view is in some way a "deterministic view" as opposed to a "stochastic" one; "stochastic" is short for "stochastic process."
3. It does not really matter whether we regard the underlying process as driven by "causes," as is customary in sociological modeling, or as driven by "choices," as is customary in economics. Logically, economists' preferences and opportunities function in a manner equivalent to "causes"; they are properties distributed among

given actors that determine outcomes. The preference model simply adds the idea that actors choose what accords with their interest in a suitably sophisticated fashion. Both sociological and economic views of causality regard the fundamental determinants of behavior as external to the actor in some sense. In the sociological case, the determinants are reified "causes" such as "race," "education," and "power." In the economic case, the determinants are the opportunities/constraints that set possible choices and the preferences that determine which of the choices is optimal. The individual per se simply acts as a locus where the intersection of these determinants takes place. For further discussion, see Abbott, A. What do cases do? Paper presented at the Northwestern University Conference on What is a case, March 1, 1990.

4. But these writers continue to at least regard different sequences of events as worthy of differentiated inquiry and hence show stronger sensitivity to the narrative character of social reality than do the more classical status attainment writers. For examples, see Hogan, D. P. 1978. The variable order of events in the life course. *American Sociological Review* 43:573-86; Marini, M. M. 1987. Measuring the process of role change during the transition to adulthood. *Social Science Research* 16:1-38; and Marini, M. M., H. C. Shin, and J. Raymond. 1989. Socioeconomic consequences of the process of transition to adulthood. *Social Science Research* 18:89-135. For a similar concern in the stress literature, see Chalmers, B. F. 1981. A selective review of stress. *Current Psychological Reviews* 1:325-43.
5. I am very uncomfortable with so strong a statement. Usually, historical processes are regarded as open to chance, but somewhat constrained; then there will be typical patterns, yet no determination *ab initio*. I reserve the term "fate" for processes that unfold in real, contingent time but that have strong teleology and consider fate to be not a scientific but a literary concept. Thus, the inevitability of tragedy arises, for Aristotle, in the original hubris of the protagonist. The details of the plot are merely the working out—the career—of that hubris. In any case, the notion that careers and other historical processes can be treated as wholes goes back to the concept of "natural history" as set forth by Robert Park and others of the Chicago school of sociology. Park on race relations; Edwards on revolutions; Shaw on delinquent careers; Thrasher on gangs; Hughes on occupations; Burgess, Reckless, Cressey, and many others on neighborhoods—all saw characteristic developmental patterns. In every case, "natural history" denoted development shaped by internal forces and environing constraints, but taking a characteristic pattern or form.
6. Abell's principal work in this vein is Comparative narratives. 1984. *Journal for the Theory of Social Behavior* 14:309-31; Analyzing qualitative sequences, in Abell and M. Proctor, eds. 1985. *Sequence analysis*. Brookfield, Vt.: Gower, 99-115; and 1987. *The syntax of social life*. Oxford, Eng.: Oxford University Press. The classic citation on optimal matching techniques is Sankoff, D., and J. B. Kruskal, eds. 1983. *Time warps, string edits, and macromolecules*. Reading, Mass.: Addison Wesley. The techniques are applied to dance sequences in Abbott, A., and J. Forrest. 1986. Optimal matching methods for historical data. *Journal of Interdisciplinary History* 16:473-96; and are also applied to careers of German musicians in Abbott, A., and A. Hrycak. Measuring resemblance in social sequences. *American Journal of Sociology*, 96:144-85. The stability of the methods under coding variation is shown in Forrest, J., and A. Abbott. The optimal matching method for anthropological data: An introduction and reliability analysis. *Journal of Quantitative Anthropology*, forthcoming. For an overview of sequence methods see Abbott, A. A primer on sequence methods. *Organization Science*, forthcoming.
7. I am not urging a turn away from positivist methods in the usual sense. Thinking about things narratively means thinking along cases rather than across them, as I have argued in Abbott, A. 1983. Sequences of social events. *Historical Methods* 16:129-47. It does not necessarily involve a turn to interpretive methods, although the two have shown an elective affinity in the past. People who have argued for or assumed a necessary connection (e.g., Richardson, L. 1990. Narrative and sociology. *Journal of Contemporary Ethnography* 19:116-35) are conflating the problem of "multiple means," (that variables can have more than one meaning in a given model, in positivist terms) and the problem of tem-

- porality proper. I am avoiding that conflation. Therefore, in this article *narrative* means narrative positivism. Even so, multiple meanings can in fact be formalized as well, as in Barthes, R. 1974. *S/Z*. New York: Hill and Wang. So even there, the assumption of "inherent unformalizability" is an error. If one does not recognize that the positivist/interpretivist dichotomy is a recursive one, using it obscures rather than enlightens. See Abbott, A. Positivism and interpretation in sociology. *Sociological Forum*, forthcoming.
8. See Abbott, A. 1988. Transcending general linear reality. *Sociological Theory* 6:169-86. That article gives a general analysis of what may be called the philosophical assumptions implicit in standard linear models. The present article elaborates the aspects of those assumptions that relate to the passage of time. For analyses of the idea of cases and variables, see Abbott, A., "Transcending"; Abell, *Syntax*; and Ragin, C. 1987. *The comparative method*. Berkeley, Calif.: University of California Press.
 9. Hence demography is the foundational science of the stochastic view. Note, however, that the tendency to sharply separate existence and other variable attributes implies a curious separation between demographic and causal models that in fact bedevils this kind of research.
 10. For an extensive analysis of this issue see A. Abbott, What do cases do?
 11. A two-dimensional analogy may help clarify the procedure. Considering a cloud of points in two-dimensional cartesian coordinates, I am looking at the smallest rectangle that could contain all the points (there will be one or more points on each edge) and then squeezing that rectangle to make it a square, one unit on a side. I am then putting graph paper of various degrees of fineness over the square and seeing how many cases are in each cell of the graph paper.
 12. Beyond the four-cut, Poisson models virtually must fit because of the extremely small parameter. Some readers may suspect, as I did, that the emptiness of the state-space is caused largely by outliers, which stretch the boundaries and create empty internal space, as scatterplots often illustrate. However, "outlierness" does not seem an issue here. (Of course, there is a large amount of literature on "regression diagnosis," which deals with issues of outliers.) In particular, I propose that "outlierness" is essentially a fractal property in many datasets; if the obvious outliers from a complex dataset are deleted, the result is a smaller dataset with equally obvious outliers. They could then be deleted, and so on.
- I demonstrate this property on the present data, as follows. I calculated a centroid at the means of each of the four variables. I then calculated the distances from each case to that point and selected the maximum distance as a standard radius. Next, I created ten concentric, four-spheres, one with a radius equal to one-tenth of that standard radius, one at two-tenths, one at three-tenths, and so on. (In two-dimensional terms, this is like drawing bull's eyes around the center of the space.) I then asked what percentage of the cases lie in the outermost spheres, these presumably being the outliers. I deleted those in the outermost spheres, recalculated the centroid, the distances, and the decile spheres, and repeated the process. The following table shows the number of cases remaining, the number in the outermost sphere, and the percentage in the three outermost spheres as I continued to go through this process of outlier deletion.
- | Remain | Drop | Sphere 8 | Sphere 9 | Sphere 10 |
|--------|------|----------|----------|-----------|
| 51 | 3 | .04 | .02 | .06 |
| 48 | 3 | .06 | .04 | .06 |
| 45 | 4 | .27 | .04 | .09 |
| 41 | 6 | .27 | .24 | .14 |
| 35 | 9 | .23 | .23 | .25 |
| 26 | 9 | .12 | .23 | .35 |
| 17 | 4 | .29 | .24 | .24 |
- A sensible definition of outlierness, given an assumption of underlying multivariate normality, would be that the outermost cases are outliers (to be dropped) if there is empty space between them and the main body of the cases. Yet in the first three situations above, there are more cases in the tenth sphere than in the ninth sphere. Deleting does not help. The data do not properly
- "tail off" until I have deleted 20 percent of the cases. The percentage of outliers does not significantly change until I have deleted a substantial portion of the data, because deleting some outliers just transforms other cases into outliers.
13. As those were the data easiest at hand, this is a cross-sectional example. The situation would be the same if I were to link up these measurements over several time periods; the state-space would be a lot emptier than we might expect. In cross-sectional data, discovering the kind of local regularity here documented is the central task of descriptive methods like clustering and scaling, which create the typologies I have suggested here. I have clustered this data using a Euclidean metric in the four-space and find a number of clusters. Sometimes the members of a cluster are likely for obvious reasons (Louisiana and Texas), sometimes for less obvious reasons (Vermont, Wisconsin, and Pennsylvania), and sometimes quite unlikely (Illinois, Nevada, Michigan, New Mexico, Maryland, and Delaware, all closer to each other than are Louisiana and Texas). Note that narrative analysis as I am proposing it here is essentially an extension of typological thinking to the temporal realm. One might further note that these results show how unrealistic the scatterplots are for the "no correlation" examples typically shown in statistics texts. In such diagrams, the points are usually well scattered throughout the space. In fact, even on Poisson assumptions, there will be considerable clumping.
 14. I am introducing an example here based on Pavalko, E. 1989. State timing of policy adoption. *American Journal of Sociology* 95:592-615. See also the analysis of that article in Abbott, What do cases do? Pavalko analyzes forty-eight real cases (states), over up to twenty-one time periods (until adoption of a compensation act), which in the world of event-history analysis serves as a data set of 369 "state years," which are the analyzed cases. (I discuss this issue of "caseness" in the article mentioned earlier.) There are seven dichotomous variables, all of which have to take on both values at some point. There are seven continuous variables. Each of them is correlated with at least one other at an absolute value of .27 or more. Correlations as large as .40 and -.67 are reported. Most of this continuous state-space is undoubtedly empty.
 15. Demonstrating this argument is the main task of my paper "What do cases do?"
 16. For my previous analysis of this problem, see "Transcending."
 17. There is a large literature on how historical arguments actually work. The classic citation on followable narratives is Gallie, W. B. 1964. *Philosophy and the historical understanding*. New York: Schocken.
 18. A serious controversy pitted Neyman against Yates on the latter's 1935 concept of "main effects" in factorial design. For a discussion, see Traxler, R. H. A Snag in the history of factorial experiments, pp. 283-95 in Owen, D. B., ed. 1976. *On the history of statistics and probability* New York: Mersel Dekker. I am here following Neyman's argument that Yates's procedure makes profound assumptions about the well-behaved character of reality. Yates's aim in adopting the concept of main effects was not, of course, to understand reality, but rather to decide whether to use a certain agricultural regimen. The main-effects approach, that is, ultimately derives from the attitude of operationalism. The same applies, of course, to the modern social science usage of main effects, which is in many cases aims at showing the positive or negative effects of various policies and personal characteristics rather than at understanding reality.
 19. Filtering is a standard approach to this issue outside social science, but generally concerns cases where one can assume an underlying pure signal and hence can make some strong assumptions in order to clean it up. Nonetheless, filtering could well see wider use in social sciences (other than economics, where it is already used widely).
 20. How we move the bell is of course a function of the signal/noise problem. I am already taking for granted here the absolute signal/noise issues discussed in an earlier paragraph.
 21. The "average" of these n situations is a scalar transform of their sum, so the waiting time till the average passes the limit set in the individual cases is the same as the waiting time till the sum passes n times that limit.
 22. Note that this time is, essentially, independent of the level we have chosen in terms of the original value of the variable; we could

- have chosen the seventy-fifth percentile or the ninety-ninth. Rather it is a function of where our criterion is located in the new, hypothetical distribution (i.e., at the mean in this case). This underlines that I have actually defined, in statistical terms, a "type 2" time horizon, one in which we are concerned with the wait to see a true change. There is also clearly a "type 1" time horizon, a waiting time until we make a false conclusion of change when actually there is none. This, too, will be larger for aggregated variables, but unlike the type 2 time horizon, it will be a function of the criterion level in the original distribution. For a formal discussion of convolutions and their application to these particular distributions, see Feller, W. 1968. *An introduction to probability theory and its applications*. New York: Wiley, 266-69. One might be tempted to think that since the aggregated variable involves n trials of the individual variables, one trial of the aggregate, which is really n trials of individuals, is the waiting time, just as in the individual case. This is not so. Each trial of the sum entails trials of *each* individual case. It is n trials of the ensemble that concern us here, which will take longer (n times longer) than the waiting time in the case of individuals. Note also that my assumption of immediate stepwise change in fact makes this a conservative figure; with continuously changing variables, the waiting time will be still longer. These facts imply deep problems in the use of summed social scales in temporal designs.
23. This problem is not escaped by event-history methods, which in their most common form—partial-likelihood models—evade the normal problems of multicollinearity that arise in the estimated parameter variances by avoiding variance-based estimation. The writing of a likelihood equation, the fundamental step in likelihood estimation, assumes the joint independence of the observations, something unfortunately impossible to assume in an event-history data array; what happened in Massachusetts in 1912 certainly is conceptually dependent on what happened there in 1911. Under certain conditions in the discrete-time estimation of event-history models, the estimation equations do reduce to a situation of apparent independence. One condition is that the vector of explanatory variables explains all variation in the hazard rate. (See P. D. Allison, "Discrete-time methods for the analysis of event histories," in S. Leinhardt [ed.], *Sociological methodology* 1982. San Francisco: Jossey-Bass.) The other condition is that of full rank for the matrix of explanatory variables, an assumption violated by any autoregression among them. Econometricians have worried about this issue (see P. Kennedy, 1985. *A guide to econometrics*. Cambridge, Mass.: MIT Press, 38), but it does not prominently feature in sociological discussions of event-history methods. I have discussed the concept of time horizons—with an example from time-series analysis—in "Transcending."
 24. One might imagine an event affecting an event already in process by a process analogous to catalysis. In that case, an event would modify an already-developing event of larger time horizon than itself. Certainly one can tell narratives this way. How to embody them in empirical practice is a different matter.
 25. For a formal analysis of these problems of colligation and measurement under the first approach, see Abbott, A. 1984. Event sequence and event duration: Colligation and measurement. *Historical Methods* 17:192-204. I have applied some of the techniques there described to sequences of welfare state development in Abbott, A., and S. DeViney, Sequences of welfare state development. Paper presented at the American Sociological Association Meeting, San Francisco, Calif., August 1989; and to sequences of development in American medical communities in Abbott, A. 1984. Professionalization large and small. Manuscript, Rutgers University. The literature on colligation is expanding. M. S. Poole, in particular, has followed up the lead of Bales's Interaction Process Analysis. See Poole, M. S., and J. Roth. 1989. Decision development in small groups IV. *Human Communication Research* 15:323-56; and 1989. Decision development in small groups V. *Human Communication Research* 15:549-89.
 26. For a detailed discussion of this "more is better" issue see Tuma, N. B., and M. Hannan. 1984. *Social dynamics*. Orlando, Fla.: Academic Press, 82-91. They argue that the practically optimal spacing of waves is one that reduces the probability of two or more changes between waves to a negligible amount. Their argument rests on the conception that I have called "time horizon," the typical waiting time for a substantial change. They do not consider the issue of "disappearance of causes" at the microlevel, which is discussed below. That is, while they consider the time horizon of the dependent variable, they do not consider those of the independent ones.
 27. The event-history framework as usually used with historical data employs annual measurement because that is the typical framework of the records. Thus, the microreduction problem described in the text does not force itself on the event historians.
 28. The relation of micro- and macrostructures is a widely discussed issue, in sociology at least. The best macrotheories are relatively institutional and treat the microlevel as relatively fixed. Marx, Weber, and Durkheim all tended to think this way, as have their current votaries. In contrast, by far the most exciting microtheories treat the microlevel as extremely open and uncertain. The Chicago School and the ecologists and symbolic interactionists who descend from them all take this position. Despite some valiant efforts, no one has really managed to put Humpty Dumpty together again. Although there is a large literature on the issue in current sociological theory, most of it lacks data, applicability, indeed even comprehensibility. It consists of "theory fixes" that stick together a few general concepts and claim to have solved the problem. Some standard views of the micro/macro relation are (1) macrostructures are simple aggregates of micro occasions, (2) micro occasions are mere instances of macrostructures, (3) micro occasions are stochastic realizations of macroprocesses, (4) macrostructures are constraints (alternatively, facilitators) for micro occasions, (5) micro is the level of freedom and macro that of constraint, (6) macrostructures are simply stochastic limits of sequences of micro occasions, and (7) microstructures are psychological, whereas macrostructures are social. None of this literature recognizes the micro/macro problem as inherently temporal, that is, none of it recognizes the conceptual, as opposed to practical, problem of time horizons. The standard review of this literature is Alexander, J. C., B. Giesen, R. Munch, and N. J. Smelser. 1987. *The micro-macro link*. Berkeley, Calif.: University of California Press.
 29. Mead's general analysis of temporality is found in *The philosophy of the present*. 1932. Chicago: University of Chicago Press. Useful summaries are Tonnes, A. 1932. A note on the problem of the past. *Journal of Philosophy*, 29:599-606; and Eames, E. R. Mead's Concept of Time. in Corti, W. R., ed. 1973. *The philosophy of George Herbert Mead*. Amrisvil, Switzerland: Amrisvil Press, 59-81. Of course, state variables such as doctoral location remain present, and so one may draw path arrows as if they had deferred effects. But the guts of the stochastic model of reality is that the entire situation at one instant produces the entire situation at the next. If part of that production is simple reproduction, then so be it. What matters is the continuity of career time and the reality of the underlying stochastic process.
 30. Mead, of course, would argue that in fact all the causality is happening in the making of the immediate present. It is merely an analytical convenience or shorthand to regard the distant past as efficacious.
 31. My current work attempts to use optimal matching to consider multiple-case, multiple-variable time-series data. This is the problem of finding different "tracks" through the state-space of variables. Citations to articles illustrating other applications of narrative positivism may be found in earlier footnotes.